

- Q.30 Write any five applications of RTD.
- Q.31 What is differential capacitive pick up? Explain its advantages.
- Q.32 How does a hot wire anemometer measure flow velocity? What are the applications of hot wire anemometer?
- Q.33 Explain the working principle of shaft encoder. Write any two disadvantages of it.
- Q.34 Explain the working of capacitive microphone with diagram.
- Q.35 Define sensor and transducer. How they are different from each other.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw the diagram of seismic transducer. Explain its working. Write any 2 advantages and disadvantages of it. What are the applications of seismic transducer?
- Q.37 Draw the diagram of RVDT. Explain the working principle of RVDT. Write any four advantages of it. Give any two differences between LVDT and RVDT.
- Q.38 What is the working principle of inductive transducer? Write any 2 advantages and disadvantages of it. How does a small displacement can be measure using inductive transducer.

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3rd Sem / IC Subject:- Transducers and Signal Conditioning

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following should not be the characteristic of a transducer?
- It should be highly linear
 - It should be accurate
 - It must be precise
 - The resolution must be very low
- Q.2 Which of the following is an active transducer?
- RTD
 - Thermistor
 - Thermocouple
 - Bourdon tube
- Q.3 An RVDT works on the principle of
- Mutual inductance
 - Self inductance
 - Resistance change
 - Capacitance change
- Q.4 A hot wire anemometer measures
- Angular velocity
 - Temperature
 - Acceleration
 - Flow velocity
- Q.5 To measure the displacement in mm range, we can use
- LVDT
 - Potentiometer

- c) Capacitive displacement transducer
- d) Spring
- Q.6 We can convert rotational displacement into voltage with the help of
 - a) LVDT b) Wheat stone bridge
 - c) RVDT d) Amplifier
- Q.7 Piezoelectric transducer is a/an
 - a) Active transducer b) Passive transducer
 - c) Inverse transducer d) Both a and c
- Q.8 Strain gauge is a/an
 - a) Resistive transducer
 - b) Inductive transducer
 - c) Capacitive transducer
 - d) Active transducer
- Q.9 RVDT stands of
 - a) Rotary variable differential transducer
 - b) Resistance variable differential transducer
 - c) Rotary variable differential transformer
 - d) Resistance variable differential transformer
- Q.10 Potentiometer converts
 - a) Displacement into electrical quantity
 - b) Flow into electrical quantity
 - c) Force into electrical quantity
 - d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Expand LVDT.

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- Q.12 Define sensor.
- Q.13 RVDT converts _____ into electrical quantity.
- Q.14 A strain gauge converts pressure into electrical quantity. (True/False)
- Q.15 What is linearity?
- Q.16 A low pass filter allow all the frequency components above the lower cut-off frequency. (True/False)
- Q.17 Thermistor is inverse transducer (True/False)
- Q.18 Generally all the transducer should have high resolution. (True/False)
- Q.19 Define Precision.
- Q.20 Define Accelerometer

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write any five characteristics of transducer.
- Q.22 Explain the working of inductive microphone. Write any two applications of it.
- Q.23 Write any five advantages of strain gauge.
- Q.24 Explain the working principle of capacitive microphone. Write any three application of it.
- Q.25 Define shaft encoder? What is the working principle of it.
- Q.26 Explain how frequency is converted into voltage.
- Q.27 What is an accelerometer? Write any four applications of it.
- Q.28 Write any five applications of RVDT.
- Q.29 Write any 5 advantages of RTD over thermistors.

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